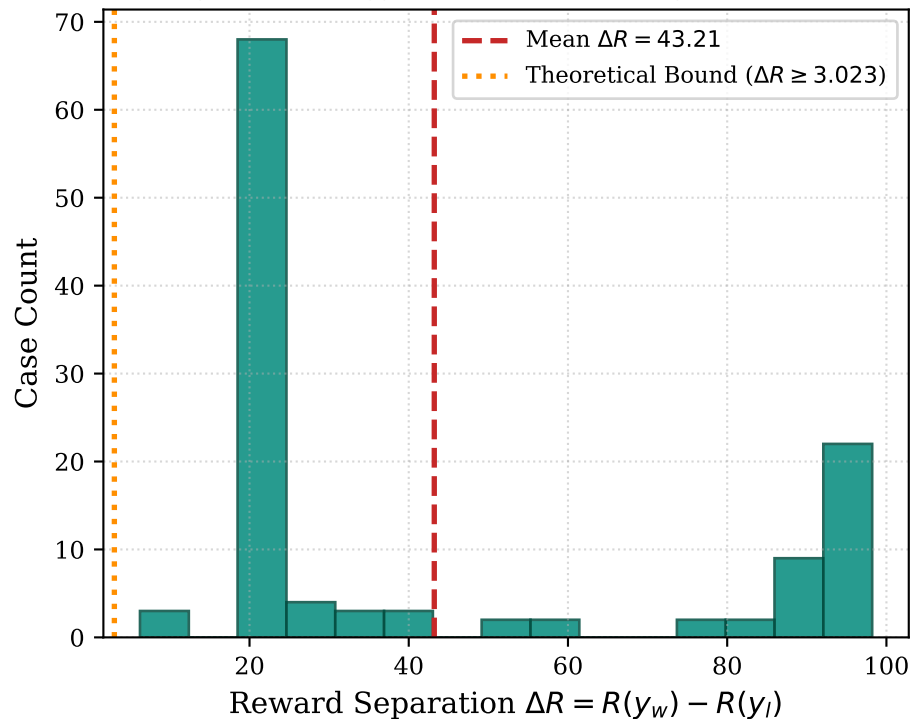


(a) Physical Energy DPO Reward Margin Distribution



(b) LoRA Fine-Tuning Convergence (-20.1% Loss)

